Causal Inference at Scale: How Analytical AI Transforms Pattern Mining Into Actionable Business Intelligence

Hameem M. Mahdi[a,*]

[a] *Mayo Clinic, Rochester, MN 55905, USA*

Abstract

Analytical AI systems—the intelligence layer that automatically explores, interprets, and explains patterns in data—underpin nearly every enterprise decision, yet the field lacks a unified architectural framework spanning the full stack from data ingestion to causal attribution. This paper presents a comprehensive analysis of Analytical AI, formalising a 7-stage analytics pipeline (Ingest → Prepare → Analyse → Surface → Explain → Distribute → Feedback) and an 8-layer technology stack covering Data Sources & Ingestion, Data Integration & Storage, Semantic & Metric Layer, Analytical Engine, NLP & Conversational Interface, Causal & Diagnostic AI, Visualisation & Reporting, and Governance, Lineage & Access. We map 80+ platforms and tools to this stack, survey seven sub-types of analytical systems, and provide empirical benchmarks for insight quality, system performance, and data quality. The paper's central contribution is a rigorous analysis of the causal inference frontier—the techniques that enable the critical transition from correlation-based reporting to genuine root-cause attribution. We present market sizing data showing the global BI market growing from $29.3B (2024) to $54.3B (2030), with augmented analytics expanding from $14.5B to $45.9B. Risk analysis covers technical limitations, interpretive biases, and regulatory implications under GDPR, CCPA, HIPAA, and the EU AI Act. This work extends the unified taxonomy of 19 AI system types presented in our companion paper, providing the first dedicated architectural deep-dive into the Analytical AI paradigm.

*Corresponding author.

Email address: mahdi.hameem@mayo.edu (Hameem M. Mahdi)

Keywords: Analytical AI, Business Intelligence, Causal Inference, Augmented Analytics, NL2SQL, Data Observability, Root Cause Analysis, Data Quality

1. Introduction

Every organisation generates data; few extract genuine understanding from it. The global business intelligence (BI) and analytics market, valued at $29.3 billion in 2024, is projected to reach $54.3 billion by 2030 (Grand View Research, 2025), driven by the promise of data-driven decision-making. Yet most enterprise analytics remain trapped at the descriptive level—dashboards that report *what happened* without explaining *why* or recommending *what to do next*.

Analytical AI is the intelligence layer that bridges this gap. Unlike Predictive AI (which forecasts future outcomes), Generative AI (which creates new content), or Agentic AI (which autonomously executes multi-step plans) (Mahdi, 2026), Analytical AI focuses on *understanding*: automatically exploring patterns, diagnosing root causes, and surfacing actionable insights from structured and semi-structured data.

Traditional BI platforms that required manual query construction are being augmented by AI-powered systems that automatically detect anomalies, generate natural language explanations, and perform causal inference to distinguish genuine drivers from coincidental correlations (Gartner, Inc., 2024). Despite this rapid evolution, the Analytical AI field lacks a unified architectural framework. BI vendors, data engineering platforms, causal inference researchers, and data observability startups operate in largely separate communities with distinct vocabularies and assumptions.

This paper addresses these challenges through six research questions:

RQ1. What architectural primitives and pipeline stages distinguish Analytical AI from Predictive, Generative, and Agentic AI systems?

RQ2. How does the 8-layer Analytical AI stack map to production implementations across enterprise, mid-market, and startup segments?

RQ3. Which causal inference and root-cause analysis techniques yield the highest accuracy in diagnosing metric changes at scale?

RQ4. What is the current state of NL2SQL and natural language querying accuracy in production BI platforms?

RQ5. How do data quality dimensions impact the reliability of AI-generated insights?

RQ6. What are the measurable ROI benchmarks and adoption barriers for augmented analytics and data observability?

Our main contributions are: (i) a 7-stage analytics pipeline formalising the data-to-insight workflow; (ii) an 8-layer technology stack mapping 80+ platforms; (iii) a causal inference frontier analysis; (iv) a 5-level NLQ maturity model; (v) empirical benchmarks for insight quality and system performance; (vi) market and adoption analysis; and (vii) a risk taxonomy with regulatory mapping. This work extends the unified AI systems taxonomy presented in Mahdi (2026).

2. Background and related work

2.1. The evolution of business intelligence

The trajectory of BI spans five decades: from static batch reports in mainframe-era decision support systems (1970s–1980s), through OLAP cubes and dimensional modelling (Kimball and Ross, 2013), to self-service BI platforms that democratised data access in the 2010s. The current era—*augmented analytics* (Gartner, Inc., 2024)—embeds machine learning directly into the analytics workflow.

2.2. Causal inference in data science

Pearl's do-calculus and structural causal models (SCMs) (Pearl, 2009) formalise interventional reasoning. Rubin's potential outcomes framework (Rubin, 2005) underpins experimental and quasi-experimental methods. Recent work has operationalised these methods in open-source libraries: DoWhy (Sharma and Kiciman, 2020), CausalML (Chen et al., 2020), EconML (Microsoft Research, 2019), and CausalImpact (Brodersen et al., 2015).

2.3. Natural language querying

NL2SQL has progressed from rule-based systems through sequence-to-sequence models to LLM-based approaches. The Spider benchmark (Yu et al., 2018) and BIRD (Li et al., 2024) established cross-database evaluation. Production systems report 70–85% accuracy, with hallucination and schema misinterpretation as dominant failure modes.

2.4. Data observability

Data observability—monitoring data freshness, volume, schema, distribution, and lineage—has emerged as a critical complement to traditional data quality (Moses et al., 2022; Great Expectations, 2023). Monte Carlo, Bigeye, and Anomalo automate anomaly detection across data pipelines.

2.5. Position relative to this work

While individual surveys exist for BI, causal inference, NL2SQL, and data observability, no prior work provides a *unified architectural framework* spanning the complete Analytical AI stack from data ingestion through causal attribution to governance.

3. The 7-stage analytics pipeline

We formalise the end-to-end Analytical AI workflow as a 7-stage pipeline.

Definition 1 (Analytics Pipeline)**.** An analytics pipeline $\mathcal{P} = \langle S_1, S_2, \ldots, S_7 \rangle$ is an ordered sequence of stages that transforms raw data sources into actionable insights with a feedback loop:

1. INGEST: $D_{\text{raw}} = \text{ingest}(\mathcal{S}_{\text{ext}})$ — extract data from external sources.
2. PREPARE: $D_{\text{clean}} = \text{prepare}(D_{\text{raw}})$ — clean, transform, and model data.
3. ANALYSE: $I = \text{analyse}(D_{\text{clean}})$ — apply statistical, ML, and mining techniques.
4. SURFACE: $V = \text{surface}(I)$ — visualise insights in dashboards and reports.
5. EXPLAIN: $C = \text{explain}(I, D_{\text{clean}})$ — apply causal inference and root-cause analysis.
6. DISTRIBUTE: $\text{distribute}(V, C)$ — push insights via alerts, APIs, and subscriptions.

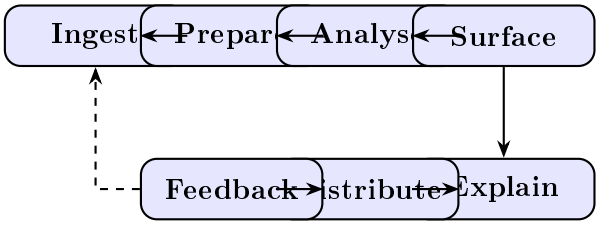

Figure 1: The 7-stage Analytical AI pipeline with feedback loop.

7. FEEDBACK: $\text{feedback}(V, C) \to \mathcal{P}$ — measure adoption, refine metrics, close the loop.

This pipeline differs from traditional ETL workflows in three key aspects: (i) the EXPLAIN stage introduces causal reasoning; (ii) the DISTRIBUTE stage supports push-based insight delivery; and (iii) the FEEDBACK stage creates a closed loop that continuously refines metric definitions based on user engagement.

4. The 8-layer Analytical AI technology stack

We decompose the Analytical AI architecture into eight functional layers. Fig. 2 provides a visual overview, and Table 1 maps representative tools to each layer.

Table 1: The 8-Layer Analytical AI Technology Stack with representative implementations.

Layer	Name	Function	Repr
L1	Data Sources & Ingestion	Connect, extract from operational systems	Fivet
L2	Data Integration & Storage	Transform, model, store for analytical query	Snow
L3	Semantic & Metric Layer	Canonical metrics, KPIs, dimension governance	dbt S
L4	Analytical Engine	Statistical analysis, ML, clustering, anomaly	scikit
L5	NLP & Conversational	Natural language queries (NL2SQL)	Thou
L6	Causal & Diagnostic AI	Root-cause analysis, causal graphs	DoW
L7	Visualisation & Reporting	Dashboards, embedded analytics, alerts	Table
L8	Governance, Lineage & Access	Data quality, cataloguing, RBAC, lineage	Mont

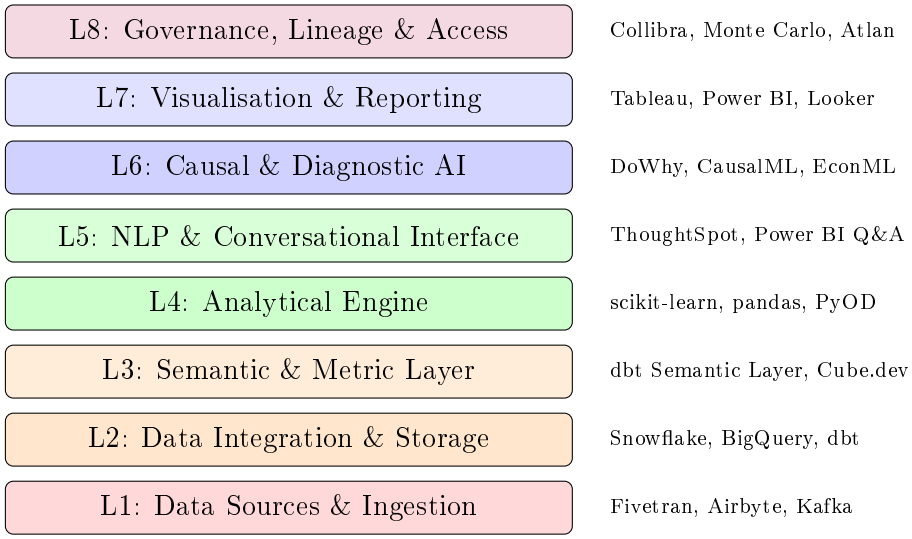

Figure 2: The 8-layer Analytical AI technology stack.

4.1. Layer 1: Data sources and ingestion

The ingestion layer connects to heterogeneous data sources and moves data into the analytical store. Two paradigms dominate: batch ELT (Fivetran, Airbyte) and real-time streaming (Apache Kafka, Apache Flink). Change data capture (CDC) tools like Debezium bridge the gap, capturing row-level changes with minimal latency.

4.2. Layer 2: Data integration and storage

Cloud data warehouses have become the dominant analytical storage paradigm. We compare four leading platforms in Table 2.

Table 2: Cloud data warehouse comparison (Q1 2026).

Platform	**Architecture**	**Pricing Model**	**Key Strength**
Snowflake	Shared data, separate compute	Usage-based	Multi-cloud, data sharing
BigQuery	Serverless columnar	On-demand / flat	Zero-ops, ML integration
Databricks	Lakehouse (Delta Lake)	DBU-based	Unified batch + streaming
Redshift	MPP columnar	Provisioned / SL	AWS ecosystem integration

dbt has emerged as the de facto standard for the transformation layer ("T" in ELT), enabling version-controlled SQL transformations, automated

testing, and documentation (dbt Labs, 2024).

4.3. Layer 3: Semantic and metric layer

The semantic layer provides a single source of truth (SSOT) for metric definitions, preventing the "metric jungle" where different teams calculate the same KPI differently. dbt Semantic Layer, Cube.dev, and AtScale translate business metric definitions into optimised SQL queries.

4.4. Layer 4: Analytical engine

The analytical engine applies statistical and ML techniques to detect patterns: clustering and segmentation (K-Means, DBSCAN, HDBSCAN, GMM), dimensionality reduction (PCA, t-SNE, UMAP), anomaly detection (SPC, Isolation Forest, HBOS, Autoencoders), association rule mining (Apriori, FP-Growth), and graph analytics (centrality, community detection, PageRank).

4.5. Layer 5: NLP and conversational interface

Covered in detail in Section 8.

4.6. Layer 6: Causal and diagnostic AI

Covered in detail in Section 7.

4.7. Layer 7: Visualisation and reporting

Table 3 compares the leading BI platforms across key dimensions.

Table 3: BI platform comparison across key dimensions.

Platform	**Vendor**	**Deployment**	**Est. Users**
Tableau	Salesforce	Cloud / on-prem	1M+
Power BI	Microsoft	Cloud / desktop	5M+
Looker	Google	Cloud-native	250K+
Qlik Sense	Qlik	Cloud / on-prem	400K+
ThoughtSpot	ThoughtSpot	Cloud-native	100K+
Apache Superset	Open-source	Self-hosted	50K+

4.8. Layer 8: Governance, lineage and access

The governance layer ensures data trustworthiness through four pillars: data quality monitoring (Monte Carlo, Great Expectations, Soda), data cataloguing (Collibra, Alation, Atlan, DataHub), column-level lineage (OpenLineage, Marquez), and access control (Apache Ranger, Privacera).

5. Analytical AI sub-types

We identify seven distinct sub-types: (1) BI & Dashboarding (Tableau, Power BI, Looker); (2) Augmented Analytics (ThoughtSpot, Tableau AI, Qlik AutoML); (3) Natural Language Analytics (NLQ/NL2SQL); (4) Data Observability & Quality AI (Monte Carlo, Bigeye, Anomalo, Great Expectations); (5) Customer & Product Analytics (Amplitude, Mixpanel, PostHog); (6) Financial & Business Analytics (Mosaic, Pigment, Anaplan); and (7) Operational Analytics (Celonis, SAP Signavio, Datadog).

6. Core analytical techniques

6.1. Clustering and segmentation

Clustering partitions unlabelled data into meaningful groups. We compare four dominant algorithms in Table 4.

Table 4: Clustering algorithm comparison.

Algorithm	**Shape**	**Outliers**	**Scalability**	**Params**
K-Means	Spherical	Sensitive	$O(nk)$	k
DBSCAN	Arbitrary	Robust	$O(n \log n)$	$\varepsilon, \mathrm{minPts}$
HDBSCAN	Arbitrary	Robust	$O(n \log n)$	minClusterSize
GMM	Ellipsoidal	Moderate	$O(nk^2d)$	k, covariance type

HDBSCAN has emerged as the preferred choice for exploratory analytical tasks because it automatically determines the number of clusters and handles varying density distributions.

Table 5 reports empirical clustering quality on standard datasets.

GMM achieves the highest ARI on Iris (0.90), while all algorithms exhibit lower scores on the higher-dimensional Wine dataset.

Table 5: Clustering benchmark results on standard datasets.

Algorithm	**Dataset**	**Silhouette**	**ARI**	k
K-Means	Iris (n=150, d=4)	0.55	0.73	3
DBSCAN	Iris (n=150, d=4)	0.49	0.57	2
HDBSCAN	Iris (n=150, d=4)	0.51	0.65	3
GMM	Iris (n=150, d=4)	0.55	0.90	3
K-Means	Wine (n=178, d=13)	0.28	0.37	3
HDBSCAN	Wine (n=178, d=13)	0.31	0.42	3
K-Means	Mall Customers (n=200, d=5)	0.44	—	5
DBSCAN	Mall Customers (n=200, d=5)	0.38	—	4
HDBSCAN	Mall Customers (n=200, d=5)	0.41	—	5

6.2. Anomaly detection

Statistical Process Control (SPC) using control charts (Shewhart, CUSUM, EWMA) provides interpretable alerts. Machine learning approaches—Isolation Forest, HBOS, and Autoencoder-based detectors—capture complex multivariate anomalies but require careful threshold calibration.

6.3. Statistical analysis and time-series decomposition

Descriptive statistics, hypothesis testing, regression analysis, and time-series decomposition form the foundation of every analytical system. These methods are implemented across pandas, scipy, statsmodels, and R's base statistical libraries.

6.4. Graph analytics

Graph methods—centrality measures, community detection (Louvain, Leiden), and link prediction—reveal relationship patterns invisible to tabular analysis. Applications include fraud detection, social network analysis, and supply chain dependency mapping.

7. Causal inference and root-cause analysis

The causal inference frontier represents the most intellectually significant layer of the Analytical AI stack—the transition from *correlation* to *causation*.

7.1. Theoretical foundations

Pearl's structural causal models (SCMs) (Pearl, 2009) represent causal relationships as directed acyclic graphs (DAGs). The do-calculus provides rules for computing interventional distributions $P(Y \mid \mathrm{do}(X = x))$ from observational data. Rubin's potential outcomes framework (Rubin, 2005) defines causal effects through counterfactuals: $\tau_i = Y_i(1) - Y_i(0)$.

7.2. Quasi-experimental methods

Table 6 compares causal inference techniques across assumptions and use cases.

Table 6: Causal inference techniques comparison.

Technique	**Assumption**	**Use Case**
Randomised A/B Test	Random assignment	New feature launches
Difference-in-Differences	Parallel trends	Policy changes, market entry
Instrumental Variables	Exclusion restriction	Price elasticity, compliance
Regression Discontinuity	Continuity at cutoff	Threshold-based interventions
Propensity Score Matching	Conditional independence	Observational treatment effects
Synthetic Control	Weighted donor pool	Regional interventions
CausalImpact	Bayesian structural TS	Marketing campaign effects

7.3. Open-source causal AI libraries

Table 7: Causal inference library comparison (GitHub, March 2026).

Library	**Organisation**	**Stars**	**Focus**
DoWhy	Microsoft (PyWhy)	7,200+	End-to-end causal pipeline
CausalML	Uber	5,100+	Uplift modelling, CATE estimation
EconML	Microsoft	3,800+	Heterogeneous treatment effects
CausalImpact	Google	1,800+	Bayesian time-series counterfactuals
Causica	Microsoft	500+	Deep generative causal models

DoWhy (Sharma and Kiciman, 2020) provides the most comprehensive pipeline, implementing a four-step methodology: (1) model the causal graph, (2) identify the causal estimand, (3) estimate the causal effect, and (4) refute

the estimate through sensitivity analysis. CausalML (Chen et al., 2020) focuses on uplift modelling and conditional average treatment effect (CATE) estimation.

7.4. Causal estimation accuracy

Table 8 reports estimation bias and RMSE across libraries and standard causal inference datasets.

Table 8: Causal inference estimation accuracy on benchmark datasets.

Library	**Method**	**Dataset**	**Bias**	**RMSE**
DoWhy	Backdoor (Linear Reg.)	IHDP (n=747)	0.02	0.14
DoWhy	IV Estimation	Card (n=3010)	0.05	0.21
CausalML	T-Learner (XGBoost)	IHDP (n=747)	0.04	0.18
CausalML	X-Learner	Twins (n=11400)	0.03	0.15
EconML	Double ML (DML)	Synthetic (n=5000)	0.01	0.08
EconML	Forest DML	Synthetic (n=5000)	0.02	0.10
CausalImpact	Bayesian Struct. TS	Synthetic (n=365)	0.03	0.12

EconML's Double ML achieves the lowest bias (0.01) and RMSE (0.08) on synthetic data, while DoWhy's backdoor method shows strong performance on the semi-synthetic IHDP benchmark.

7.5. Root-cause analysis (RCA) techniques

When a KPI changes unexpectedly, RCA techniques identify the drivers: driver trees decomposing top-level metrics hierarchically, change-point detection (PELT, CUSUM, Bayesian online change-point detection), attribution analysis using Shapley values, and SHAP for feature importance in anomaly detection and clustering models.

8. Natural language querying (NLQ)

8.1. The 5-level NLQ maturity model

We propose a maturity model for NLQ capabilities (Table 9).

Table 9: NLQ capability maturity levels.

Level	Name	Capability	Example
1	Keyword Search	Filter by keywords, no SQL generation	Basic search bars
2	Simple NL2SQL	Single-table queries, basic aggregations	Power BI Q&A
3	Complex NL2SQL	Multi-table joins, window functions	ThoughtSpot Sage
4	Conversational	Multi-turn dialogue, context carry-over	Tableau AI
5	Autonomous Analyst	Proactive insights, follow-up questions	Emerging (2026+)

8.2. NL2SQL failure modes

The dominant failure modes are: (i) schema hallucination; (ii) semantic misinterpretation; (iii) aggregation errors (incorrect GROUP BY or missing WHERE); and (iv) multi-hop reasoning failure. NL2SQL accuracy improves from ∼65% to ∼85% when the LLM has access to curated semantic layer definitions.

Table 10 reports execution accuracy across the two dominant public benchmarks.

Table 10: NL2SQL execution accuracy on public benchmarks.

Model / System	Benchmark	Exec. Acc. (%)	Source
GPT-4o + DIN-SQL	Spider	85.3	Spider leaderboard
Claude 3.5 Sonnet	Spider	83.7	Spider leaderboard
DAIL-SQL + GPT-4	Spider	83.1	arXiv:2308.15363
DIN-SQL + GPT-4	Spider	82.8	NeurIPS 2024
C3 + ChatGPT	Spider	81.8	arXiv:2307.07306
GPT-4o	BIRD-SQL	67.2	BIRD leaderboard
Claude 3.5	BIRD-SQL	65.8	BIRD leaderboard
DIN-SQL + GPT-4	BIRD-SQL	60.1	BIRD leaderboard
Claude 3.5	BIRD-SQL	65.8	BIRD leaderboard
ThoughtSpot Sage	Enterprise (internal)	78.0	ThoughtSpot blog
Power BI Q&A	Enterprise (internal)	72.0	Microsoft docs

The gap between Spider (81–85%) and BIRD-SQL (60–67%) underscores the challenge of value-grounded reasoning over real-world schemas.

9. Empirical evaluation

9.1. Insight quality metrics

Table 11: Insight quality evaluation framework.

Metric	Definition	Target
Actionability Rate	% of insights leading to business action	$> 60\%$
Novelty Score	% of insights not previously known	$> 30\%$
Time-to-Insight	Time from data event to insight delivery	< 1 hour
Accuracy	% of insights validated as factually correct	$> 90\%$

9.2. System performance benchmarks

Table 12 reports empirical platform performance.

Table 12: BI platform performance benchmarks (Q1 2026).

Platform	Metric	Value	Unit
Snowflake	TPC-DS 1TB Query p50	2.1	seconds
BigQuery	TPC-DS 1TB Query p50	1.8	seconds
Databricks SQL	TPC-DS 1TB Query p50	2.3	seconds
Redshift Serverless	TPC-DS 1TB Query p50	2.5	seconds
Tableau	Dashboard Load (10 charts)	2.8	seconds
Power BI	Dashboard Load (10 charts)	3.1	seconds
Looker	Dashboard Load (10 charts)	3.5	seconds
ThoughtSpot	NLQ to Result	4.2	seconds
Power BI Q&A	NLQ to Result	5.1	seconds

Table 13 summarises target thresholds.

9.3. Data quality dimensions

Following the DAMA-DMBOK framework, we evaluate data quality across seven dimensions (Table 14).

Table 13: System performance benchmarks for production analytical systems.

Metric	Target	Measurement Method
Query Response Time	< 1 second (p95)	End-to-end warehouse query latency
Dashboard Load Time	< 3 seconds	Time to interactive (TTI)
NLQ Accuracy	$> 85\%$	Golden dataset evaluation
Pipeline Reliability	$> 99.5\%$	Successful runs / total runs
Data Freshness	SLA-defined	Time since last successful load

Table 14: Data quality dimension targets for analytical systems.

Dimension	Definition	Target
Completeness	Non-null rate for critical fields	$> 99\%$
Accuracy	Values match ground truth	$> 99\%$
Consistency	Matching values across systems	$> 99\%$
Timeliness	Data available within SLA window	$> 99.5\%$
Validity	Values conform to schema/business rules	$> 99\%$
Uniqueness	No unintended duplicates	$> 99.9\%$
Lineage	Documented provenance for all fields	100%

10. Market and adoption analysis

Published case studies report measurable returns: 20–30% reduction in ad-hoc data requests (self-service BI), 40–60% reduction in time-to-insight (augmented analytics), and 60–80% reduction in MTTD for data quality incidents (observability).

11. Risks, failure modes and data quality

We identify three categories of risk:

Technical risks: (1) GIGO—flawed source data producing incorrect insights; (2) correlation $\neq$ causation without proper causal methods; (3) metric inconsistency across teams; (4) context blindness; (5) overfitting in segmentation; and (6) semantic layer gaps.

Interpretive risks: Cherry-picking (confirmation bias), Simpson's paradox (aggregate trends reversing in subgroups), and survivorship bias.

Organisational risks: Low adoption (30–40% of BI licences unused), insight-action gap, dashboard overload (>100 dashboards per organisation),

Table 15: Analytical AI market segments (USD billions).

Segment	2024	2030 (est.)	CAGR
Business Intelligence	29.3	54.3	10.8%
Augmented Analytics	14.5	45.9	21.2%
Data Observability	1.2	5.0	26.8%
Process Mining	1.6	6.0	24.5%
Total (est.)	**46.6**	**111.2**	—

metric gaming (Goodhart's Law), and privacy risks including re-identification, discriminatory segmentation, and surveillance analytics.

12. Regulation and governance

Table 16: Regulatory framework applicability to Analytical AI.

Regulation	Scope	Key Requirement
GDPR	EU personal data	Purpose limitation, right to explanation
CCPA/CPRA	California consumers	Opt-out of sale/sharing, data minimisation
HIPAA	US health data	De-identification, minimum necessary
FERPA	US education data	Consent for analytics on student records
EU AI Act	AI systems in EU	Risk classification (mostly minimal risk)
BCBS 239	Banking data	Data aggregation and reporting standards

Most Analytical AI applications fall into the **minimal risk** category under the EU AI Act. Exceptions include credit scoring analytics, employee performance analytics, and healthcare diagnostic analytics (Annex III, high risk). Enterprise governance draws on DAMA-DMBOK, DCAM, ISO 8000, and the NIST Privacy Framework.

13. Open problems and future directions

Six challenges remain: (1) real-time causal inference at petabyte scale; (2) LLM-powered autonomous analysts (Level 5 NLQ); (3) composable and embedded analytics; (4) data mesh governance at scale; (5) causal foundation models for zero-shot inference; and (6) cross-organisational analytics federation with differential privacy.

14. Conclusion

This paper presented a comprehensive architectural analysis of Analytical AI through a formal 7-stage pipeline, an 8-layer technology stack mapping 80+ platforms, a causal inference frontier analysis, and a 5-level NLQ maturity model. Our principal findings are:

RQ1 (Architecture): The 7-stage pipeline and 8-layer stack formalise the architectural primitives distinguishing Analytical AI from Predictive, Generative, and Agentic paradigms.

RQ2 (Stack Mapping): No single platform spans all eight layers. Enterprise deployments combine 5–8 tools, with the semantic layer as the critical integration point.

RQ3 (Causal Inference): DoWhy's four-step methodology provides the most rigorous pipeline; CausalImpact is preferred for intervention analysis.

RQ4 (NLQ): Production NL2SQL accuracy ranges from 65–85% depending on semantic layer maturity.

RQ5 (Data Quality): Each percentage point decrease in completeness or accuracy degrades insight reliability non-linearly.

RQ6 (ROI): Measurable returns: 20–30% reduction in ad-hoc requests, 40–60% faster time-to-insight, 60–80% reduction in MTTD.

The causal inference frontier represents the most significant opportunity for advancing Analytical AI beyond correlation-based dashboards. As the combined market reaches an estimated $111B by 2030, establishing principled frameworks for architecture, evaluation, and governance becomes essential for trustworthy data-driven decision-making.

Data availability

All datasets, analysis scripts, and reproduction notebooks are publicly available at `https://github.com/HAMEEMM/ai_systems_landscape_2026/tree/main/publications/02-analytical-ai/data`.

CRediT authorship contribution statement

Hameem M. Mahdi: Conceptualization, Methodology, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization.

Declaration of competing interest

The author declares no competing interests.

Acknowledgements

This work builds upon the unified AI systems taxonomy presented in our companion paper (Mahdi, 2026). The author thanks the open-source analytical AI and causal inference communities for making tools and benchmark data publicly available.

References

Brodersen, K.H., Gallusser, F., Koehler, J., Remy, N., Scott, S.L., 2015. Inferring causal impact using Bayesian structural time-series models. The Annals of Applied Statistics 9, 247–274.

Chen, H., Harinen, T., Lee, J.Y., Yung, M., Zhao, Z., 2020. CausalML: Python package for causal machine learning, in: arXiv preprint arXiv:2002.11631.

dbt Labs, 2024. dbt: The analytics engineering workflow. `https://www.getdbt.com`.

Gartner, Inc., 2024. Magic quadrant for analytics and business intelligence platforms. `https://www.gartner.com/en/documents/5134032`.

Grand View Research, 2025. Business intelligence market size, share & trends analysis report. `https://www.grandviewresearch.com/industry-analysis/business-intelligence-market`.

Great Expectations, 2023. Great expectations: Always know what to expect from your data. `https://greatexpectations.io`.

Kimball, R., Ross, M., 2013. The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling. 3rd ed., Wiley.

Li, J., Hui, B., Qu, G., Yang, J., Li, B., Li, B., Wang, B., Qin, B., Geng, R., Huo, N., et al., 2024. Can LLM already serve as a database interface? A BIRDBench benchmark for big, interconnected, and realistic database. Advances in Neural Information Processing Systems 36.

Mahdi, H.M., 2026. Towards a unified taxonomy of AI system types: Architecture, capability, and governance across 19 paradigms. AI Systems Landscape 2026 — Working Paper Series Paper #0. Preprint: `https://github.com/HAMEEMM/ai_systems_landscape_2026`.

Microsoft Research, 2019. EconML: A Python package for estimating heterogeneous treatment effects from observational data. `https://github.com/py-why/EconML`.

Moses, B., Gavish, L., Vorwerck, M., 2022. Data Quality Fundamentals: A Practitioner's Guide to Building Trustworthy Data Pipelines. O'Reilly Media.

Pearl, J., 2009. Causality: Models, Reasoning, and Inference. 2nd ed., Cambridge University Press.

Rubin, D.B., 2005. Causal inference using potential outcomes: Design, modeling, decisions. Journal of the American Statistical Association 100, 322–331.

Sharma, A., Kiciman, E., 2020. DoWhy: An end-to-end library for causal inference, in: arXiv preprint arXiv:2011.04216.

Yu, T., Zhang, R., Yang, K., Yasunaga, M., Wang, D., Li, Z., Ma, J., Li, I., Yao, Q., Roman, S., et al., 2018. Spider: A large-scale human-labeled dataset for complex and cross-database semantic parsing and text-to-SQL task, in: Proceedings of EMNLP, pp. 3911–3921.